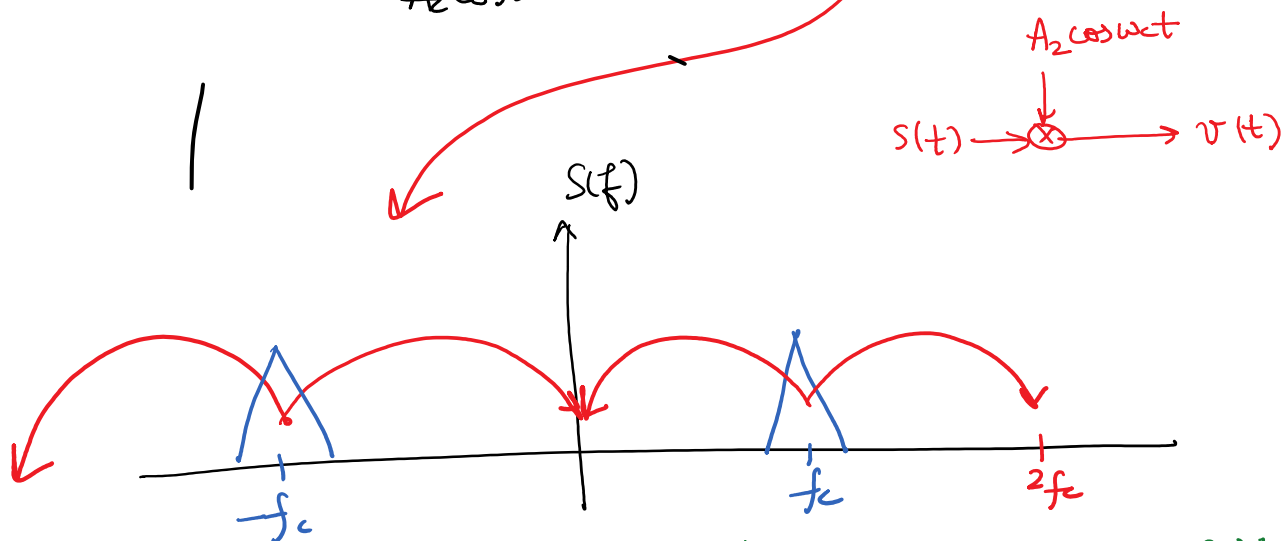
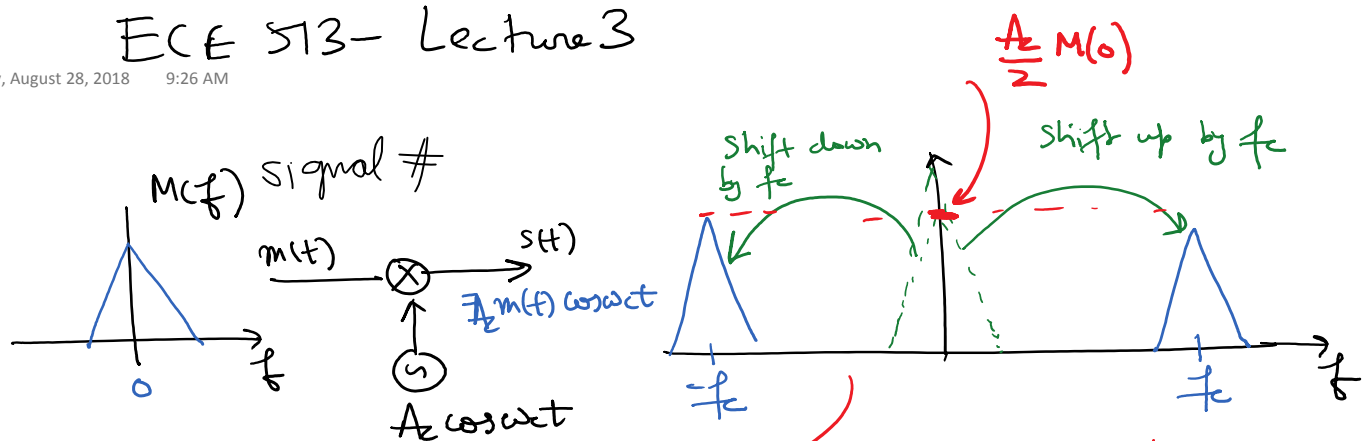
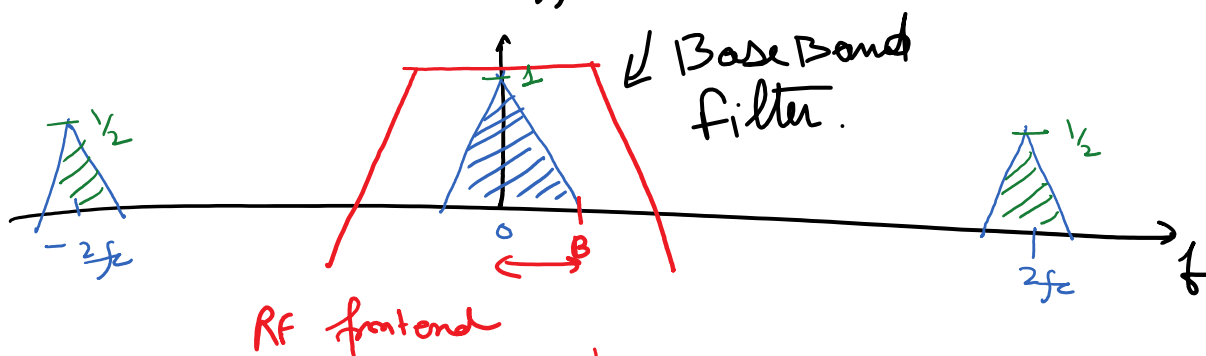


ECE 513- Lecture 3

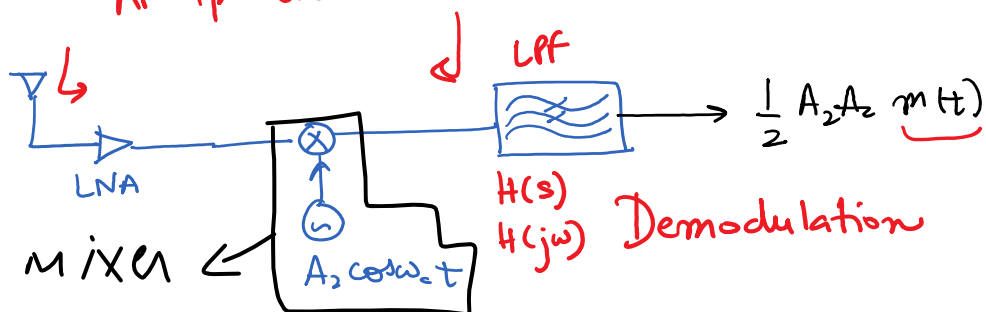
Tuesday, August 28, 2018 9:26 AM

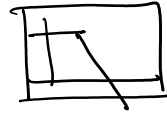


$$v(t) = \frac{1}{2} A_2 A_2 [\underbrace{m(t)}_{V(f)} + \underbrace{m(t) \cos(2\omega_c t)}_{V(f)}] \quad \rightarrow \frac{1}{2} [e^{j2\omega_c t} + e^{-j2\omega_c t}]$$



Rx





lowpass filter

In general

in-phase quadrature phase

$$m(t) = m_I(t) + j m_Q(t) \leftarrow \text{Complex baseband signal}$$

Let $A_c = 1$

transmitted signal

$$S(t) = \text{Re} \left\{ m(t) e^{j\omega_c t} \right\} \leftarrow \text{Complex modulation}$$

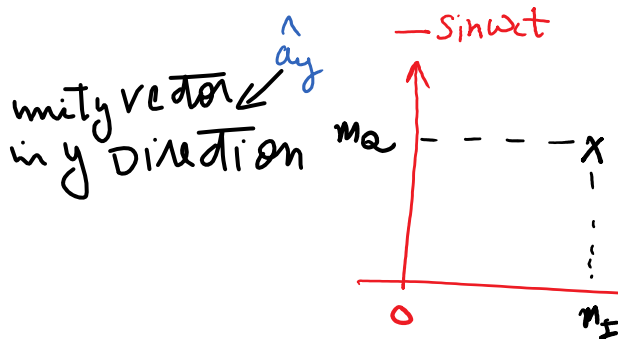
Complex carrier $\Rightarrow \cos\omega_c t + j \sin\omega_c t$

$$= \text{Re} \left\{ [m_I(t) + j m_Q(t)] [\cos\omega_c t + j \sin\omega_c t] \right\}$$

$$S(t) = m_I(t) \cos\omega_c t - m_Q(t) \sin\omega_c t$$

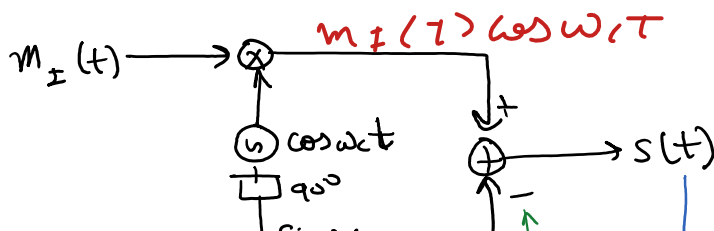
In phase

quadrature



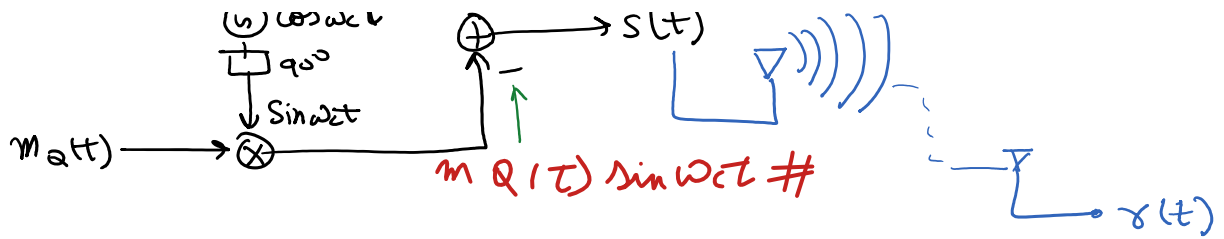
$\hat{a}_x \Rightarrow$ unity vector in X Direction

any Dot product between 2 $\perp$ comp $\Rightarrow 0$

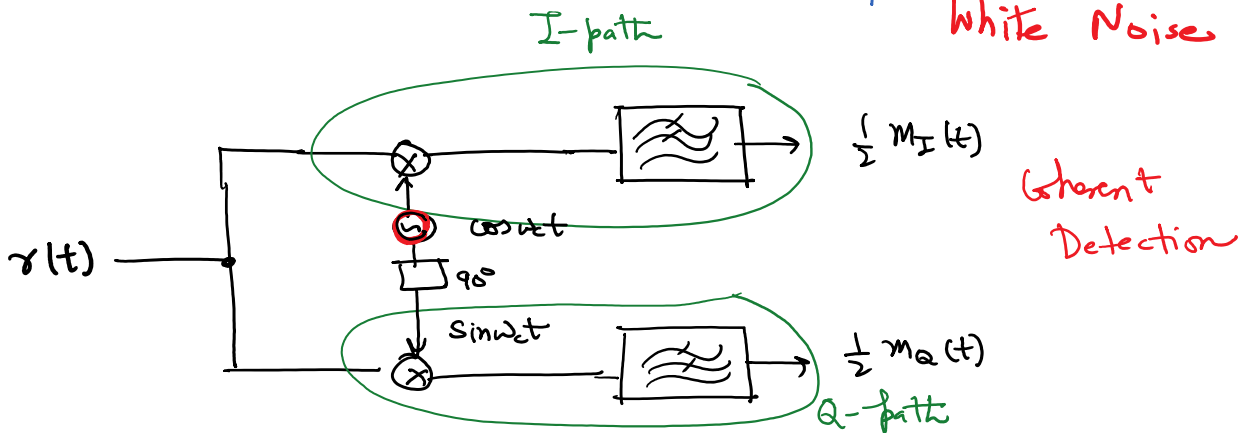
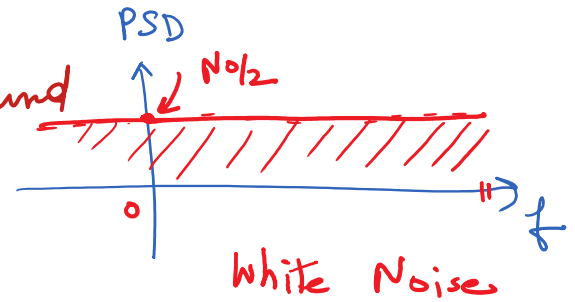
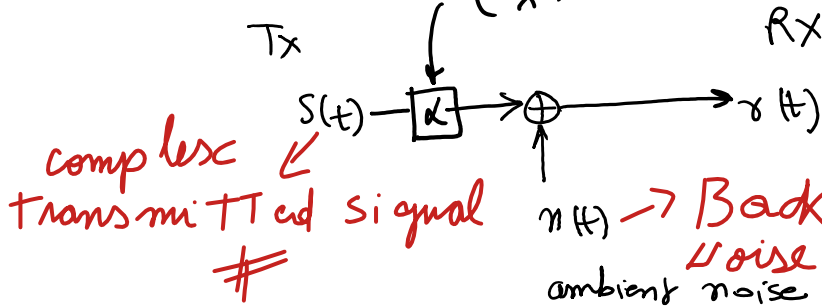
$$\hat{a}_x \hat{a}_y = \int_0^{T_b} \cos\omega_c t \sin\omega_c t dt = 0$$


$\times \cos\omega_c t + \sin\omega_c t$ form a 2D vector space





$\frac{G_T G_R}{\left(\frac{4\pi r}{\lambda}\right)^2}$ → propagation loss factor

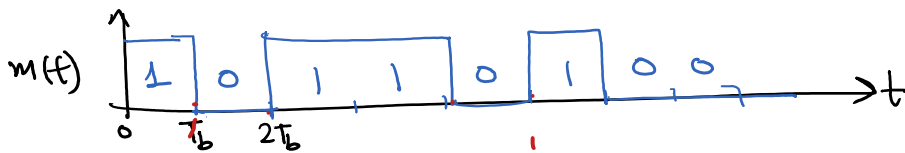


* Need the Rx oscillator to be synchronized with the Tx carrier.

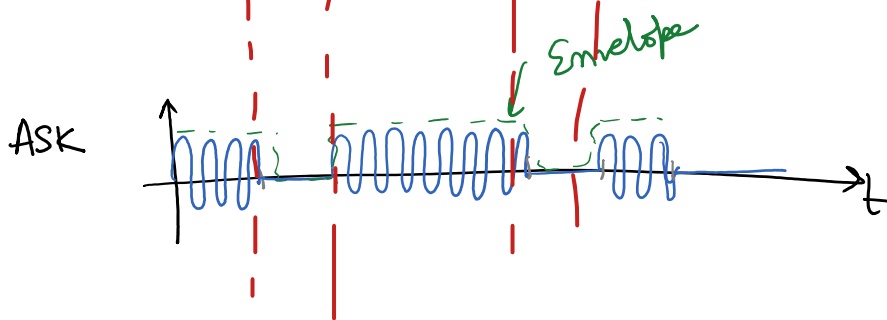
Digital Modulation :

* All modern communication systems employ digital modulation

$$T \equiv T_b \Leftarrow \text{bit period}$$



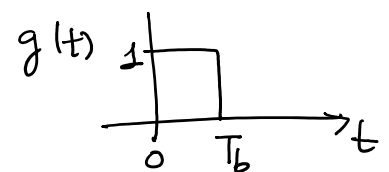
$$\text{Bit rate } R_b = \frac{1}{T_b}$$



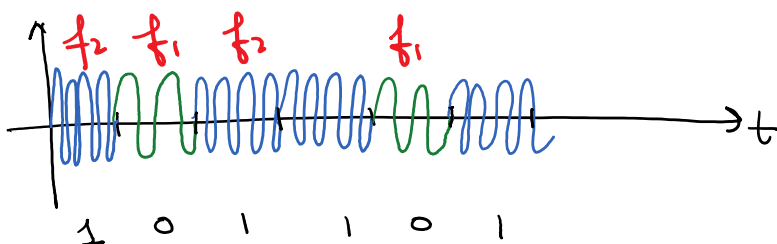
Amplitude shift keying

$$s(t) = A_c m(t) \cos \omega_c t$$

$$m(t) = \sum_{n=0}^{\infty} \underbrace{b_n}_{b_n \in \{0,1\}} \underbrace{g(t - nT_b)}_1$$



FSK → frequency shift keying



frequency offset

$$s(t) = A \cos[\omega_c t + m(t) \Delta \omega t]$$

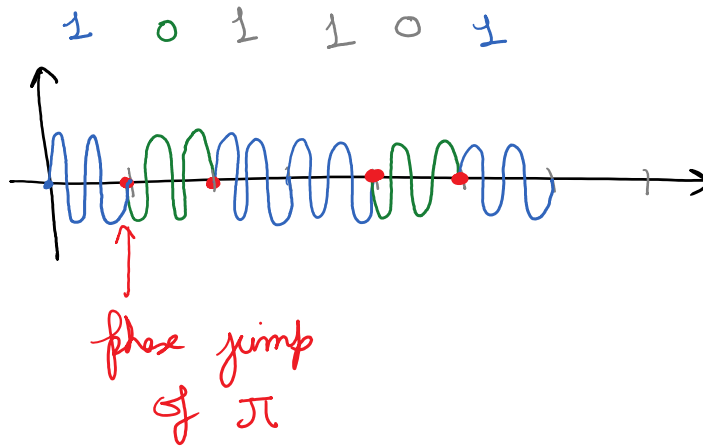
0/1

$$\Delta\omega = 2\pi \Delta f, \Delta f = f_1 - f_2$$

$$m(t)=0 \Rightarrow A_c \cos \omega_2 t$$

$$m(t)=1 \Rightarrow A_c \cos \omega_1 t$$

PSK $\rightarrow$ Phase shift keying



$$s(t) = A_c \cos(\omega_c t + \phi_n)$$

b_n	ϕ_n
0	π
1	0

Binary PSK $\Rightarrow$ BPSK

* Constant Envelope is preferred
"at times to minimize power
Amplifier distortion"

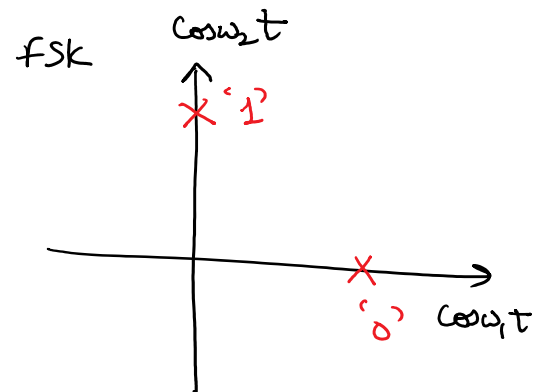
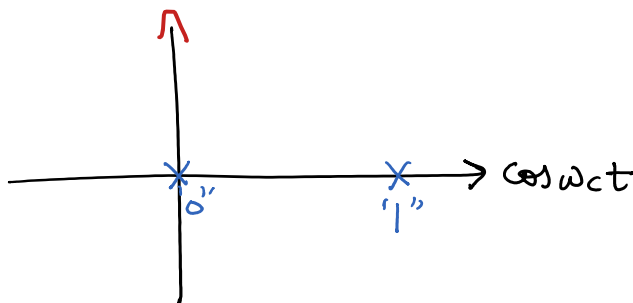
PAPR $\Rightarrow$ Peak to average power
ratio of $s(t)$

M-ary Digital Modulation Schemes.

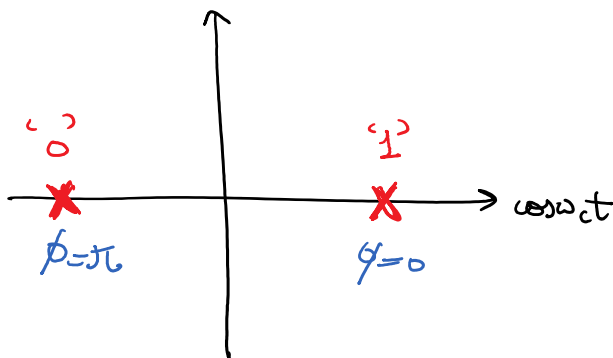
- * Binary modulation (ASK, FSK, BPSK) have a spectral efficiency of 1 bps/Hz
- * M-ary modulation packs more bits in one time period (or more bits per symbol)
 $\rightarrow$ higher data rates transmitted

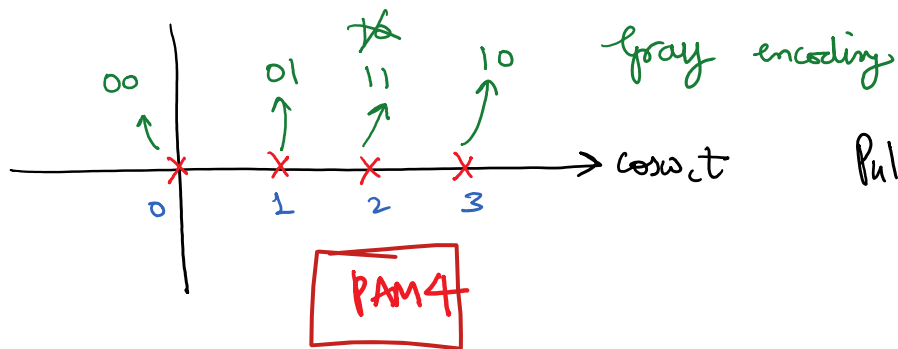
Constellation Diagram

ASK



PSK





Pulse amplitude modulation

PAM-M

$$\text{bit rate} = \log_2 M \cdot \frac{1}{T_b}$$

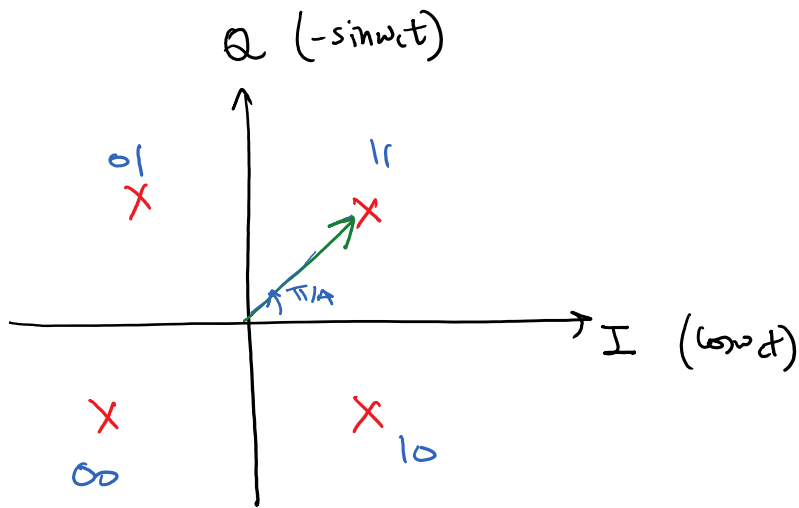
$\log_2 M$ bits per symbol

$$M=4 \Rightarrow 2 \text{ bits/symbol}$$

$$M=16 \Rightarrow 4 \text{ bits/symbol}$$

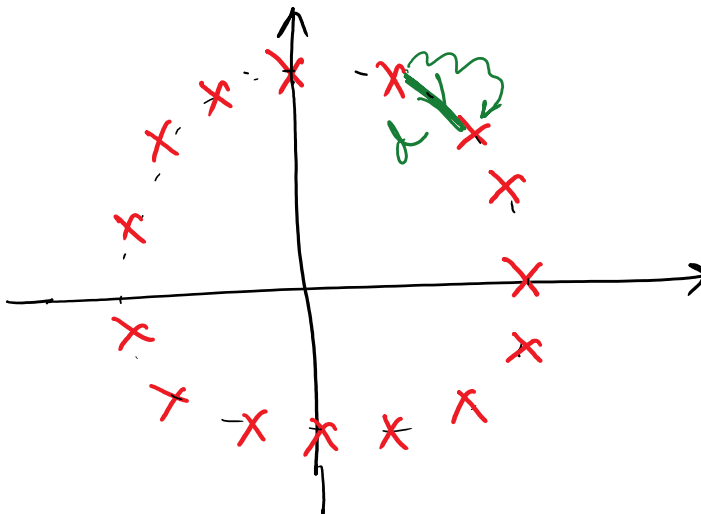
QPSK $\Rightarrow$ Quadrature PSK

$M=4$



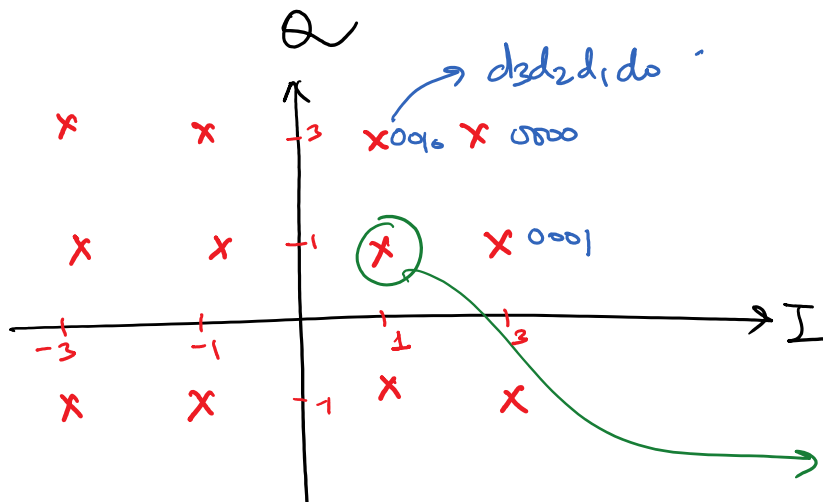
Constant Envelope

M-PSK



θ to be maximized

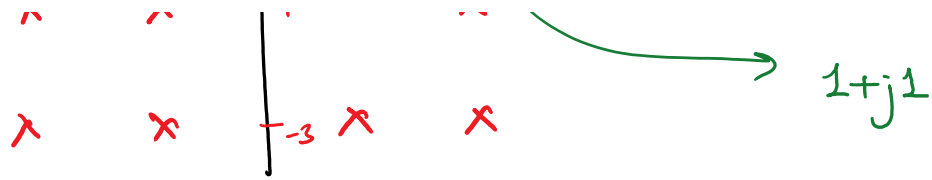
M-QAM $\rightarrow$ Quadrature Amplitude Modulation



16-QAM

4-bits/symbol

$1+j1$



$$S(t) = a_I \cos \omega_c t - b_Q \sin \omega_c t$$

$$a_I \Rightarrow \pm 1, \pm 3, \dots \pm (\log_2 M - 1)$$

$$b_Q \Rightarrow \pm 1, \pm 3, \dots \pm (\log_2 M - 1)$$

$$s(t) = \operatorname{Re} \{ m(t) e^{j\omega_c t} \}$$